

QGreenFleet — Your Fleet Plan

Date: 2026-09-30 | Fleet: 20 ships, 5 routes | Report ID: QGF-20260930-21F4F82E

The Bottom Line

- Save \$1,855,362 on fuel compared with today's plan (−60.3%) — a reduction.
- Cut carbon emissions by 13,481 tonnes (−80.0%) — a reduction — about the yearly CO₂ of 2,931 cars.
- The plan meets all route demand, arrival schedules, fuel availability and carbon-intensity limits in the model.

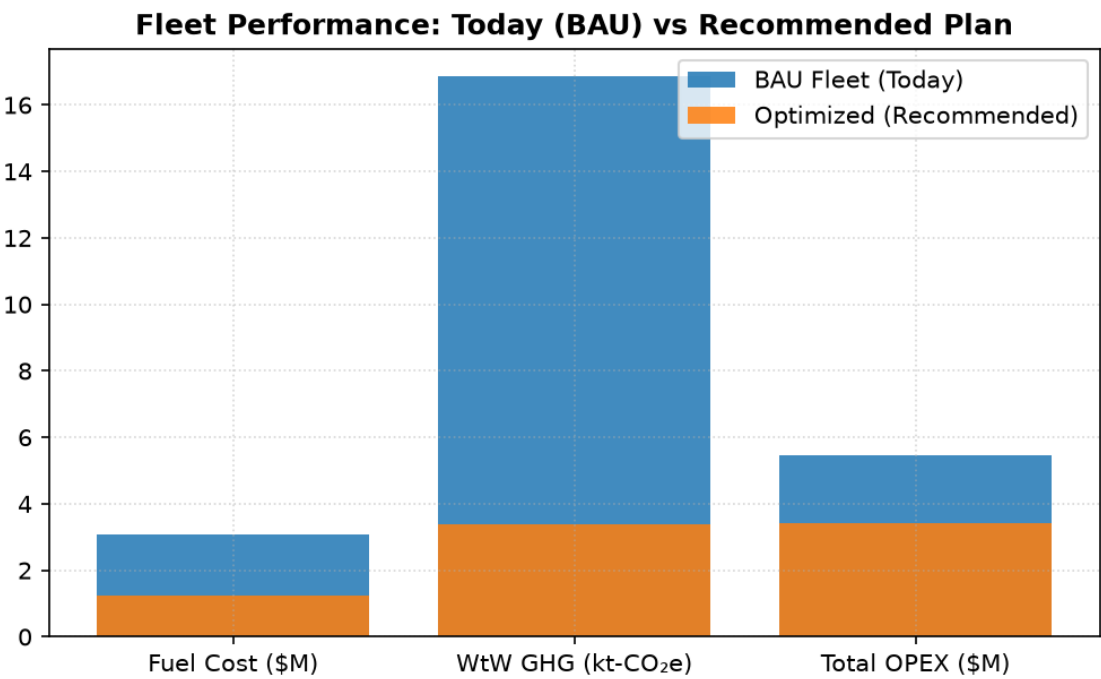


Figure 1: Today's plan (grey) vs recommended plan (green).

Where the Change Comes From

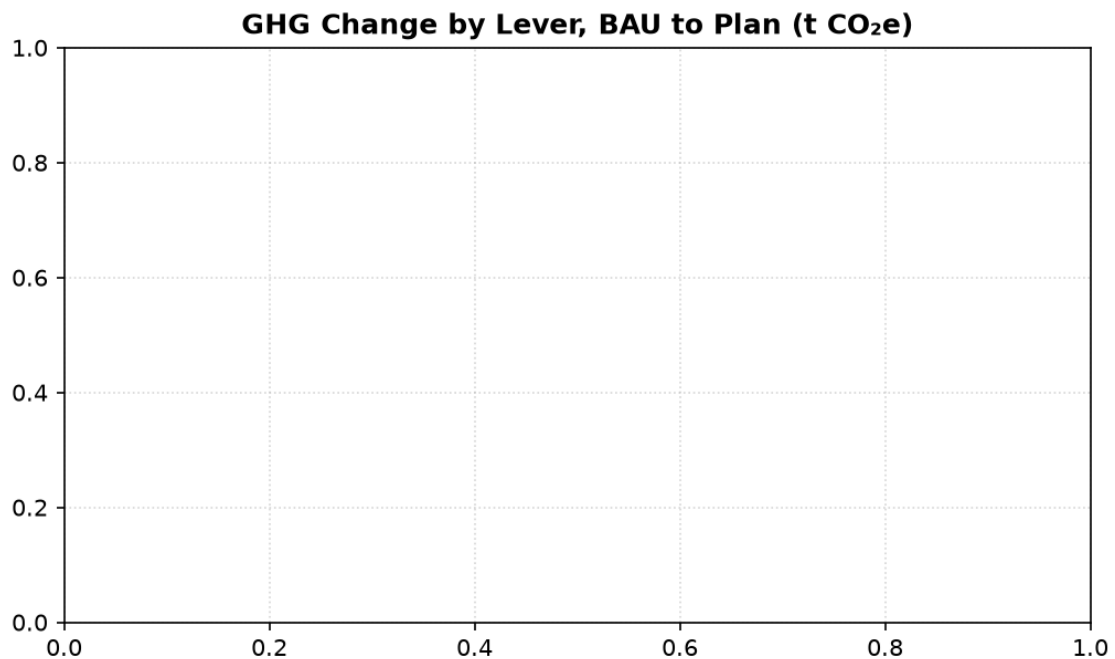


Figure 2: Change in carbon emissions from today's plan, one bar per type of change.

- ☐ **Which ships sail which routes** saves 8,071 t CO₂e.
- ☐ **Sailing speeds** saves 4,532 t CO₂e.
- ☐ **Fuel choices** saves 876 t CO₂e.
- ☐ **Plugging into port electricity** saves 3 t CO₂e.

Pick Your Priority

Option	Fuel Cost	Carbon	Extra Cost vs Cheapest	Recommended When
☐ Cheapest	\$0.83M	4,267 t	+\$0.00M	Tight budgets
☐ Recommended	\$1.22M	3,370 t	+\$0.39M	Balanced
☐ Greenest	\$3.27M	2,279 t	+\$2.44M	Emission targets

The recommended option closes 45% of the carbon gap between the cheapest and the greenest option for 16% of the extra fuel cost.

Top Ship Changes

Ship	Route	Change vs today
V000	Reserve	held in reserve (deployed in BAU)
V001	Reserve	held in reserve (deployed in BAU)
V002	Reserve	held in reserve (deployed in BAU)
V003	R0	slowed 3.8 kn
V004	Reserve	held in reserve (deployed in BAU)

Can You Trust These Numbers?

- ✓ Fuel model trained on **21,622 real ship reports** from the EU MRV register. On ships it had not seen, it explains 52% of the variation in fuel use, with a typical error of 26%.
- ✓ Official IMO and EU emission conversion factors are applied throughout.
- ✓ Carbon-intensity rating (approximate, A-E): 5/5 deployed in A-C.

How our method compares

Fleet tested	Our optimizer (run time)	Standard method (run time)	Our speed
5 ships	9.1 s	11.7 s	1.29× faster
20 ships	19.5 s	24.8 s	1.27× faster
50 ships	32.4 s	36.7 s	1.13× faster
100 ships	65.2 s	57.8 s	1.13× slower

Plan quality score: our optimizer scored best on 1 of 4 fleet sizes tested (5 random seeds each).

What to Watch

□ **Carbon price:** in the carbon-price test (\$0–\$200 per tonne), green methanol never overtook heavy fuel oil — no crossover in the swept range.